

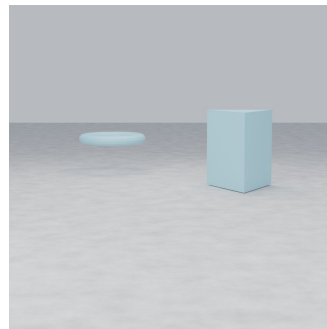
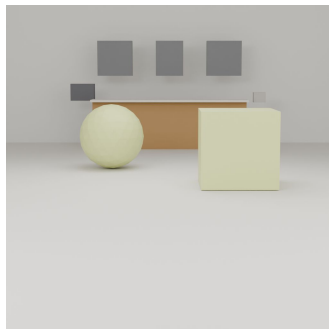
GRAFIQ

Do Vision-Language Models Understand Image Formation?

We built a scalable, verifiable VQA benchmark for VLM graphic reasoning and generation.

Steve Chen **Patrick Tsung-Han Wu** **Fei Gao** **Alex Wang**

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Sample Blender renders from the GRAFIQ benchmark.



Background and Introduction

VLMs crush lots of visual reasoning tasks — so what about graphics?

STEP 1

Object Recognition

What is in the scene?

STEP 2

Graphic Recognition

Tell depth, roughness, normal,
and lighting differences.

STEP 3

Inverse Rendering

Write Blender code
to reconstruct the scene.

GRAFIQ targets steps 2 and 3. Existing benchmarks measure step 1 well — but graphic-level reasoning and code-level reconstruction remain largely untested.

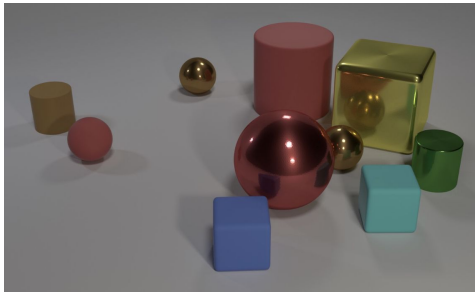


Our Contribution

1. Defined a set of pioneering graphic recognition / rendering tasks for VLMs.
2. Built a scalable, verifiable benchmark across five practical axes.
3. Evaluated seven widely-used VLMs and highlighted insights for future work.

PRIOR WORK

Object Recognition



Q. *How many objects are there in the scene?*

OURS

Graphic Recognition and Generation

Q1. *Which object has the shiniest surface?*

Q2. *Please reconstruct the scene with Blender 4.4.3...*

[1] Johnson et al. CLEVR: A diagnostic dataset for compositional language and elementary visual reasoning. CVPR 2017.

[2] Fu et al. BLINK: Multimodal large language models can see but not perceive. ECCV 2024.



Live Demo

A walk-through of GRAFIQ benchmark items.



Benchmark Categories

Five tasks: Four perception, one inverse generation.

Category	What it tests	Example question
Depth Reasoning	Identify which object is closer to the camera	<i>“Which object is closer to the camera?”</i>
Material Reasoning	Compare object roughness and glossiness	<i>“Which object has the shinier surface?”</i>
Surface Orientation	Tell which object faces the camera more directly	<i>“Which object is more front-facing?”</i>
Lighting Reasoning	Identify which object is more directly illuminated	<i>“Which object is more brightly lit?”</i>
Inverse Generation	Generate Blender code for a given scene	<i>“Generate this scene using Blender code...”</i>

Table 1. Five categories cover the principal axes of forward and inverse image formation.



Towards Scalable, Verifiable Benchmark Generation

Stage 1 Scene Generation (Blender)

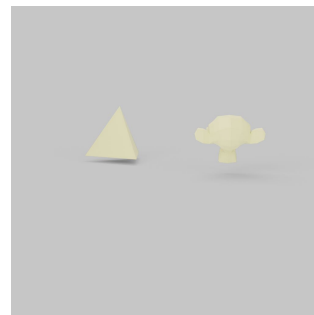
- Randomly select object pairs from Objaverse [3] dataset.
- Randomly select a scene background.
- Place objects to the scene with ONE varied factor.
- Image rendering.

Stage 2 QA Generation and Verification

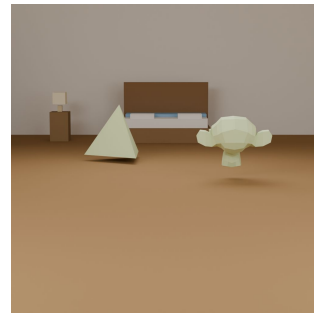
- Create a prompt for each scene (left and right matters!)
- Filter out items that might have ambiguity.

Result. We end up with **480** (120×4) synthetic samples in total.

[3] Deitke et al. Objaverse: A universe of annotated 3D objects. CVPR 2023.



Primitive Studio



Indoor Scene



Results — TL;DR

Top-line headlines from evaluating 7+ VLMs on the GRAFIQ benchmark.

01 7+ proprietary and open-source VLMs evaluated.

GPT-4o-mini, GPT-5-mini, Gemini 3.1 Pro, Claude Opus 4.5, Qwen 2.5/3-VL, GLM-4.5V.

02 Recognition $\neq$ Inverse generation.

Model rankings change drastically depending on which axis you measure.

03 Recognition: roughness solved, the rest still hard.

Depth, surface normals, and lighting still leave significant headroom.

04 Inverse generation: most models can't even produce runnable code.

Executable rate ranges from 14 % to 100 %, with frontier models far ahead.

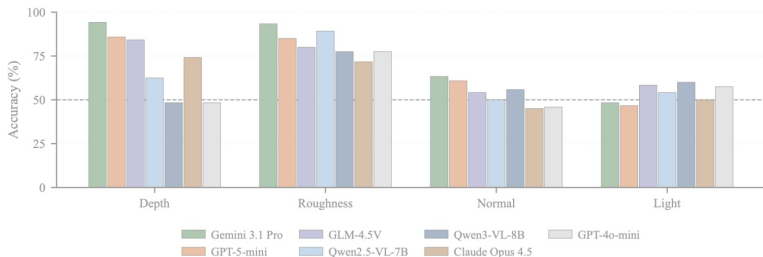


Results 1 — Recognition

§1 Breakdown Analysis

Table 2: Recognition accuracy on the $n=480$ split. Bold = above binomial chance at $p < 0.05$ vs. 50%. All = overall accuracy with 95% bootstrap CI; D/R/N/L = per-task % for depth, roughness, normal, light ($n=120$ each).

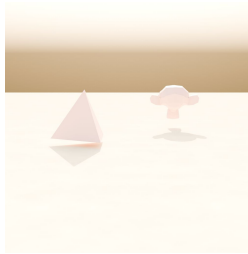
Model	All	D	R	N	L
Qwen2.5-VL-7B-Instruct	64.0% [59.6, 68.3]	62.5%	89.2%	50.0%	54.2%
Qwen3-VL-8B-Thinking	60.4% [56.0, 64.8]	48.3%	77.5%	55.8%	60.0%
GLM-4.5V-Thinking	69.2% [65.0, 73.3]	84.2%	80.0%	54.2%	58.3%
GPT-4o-mini	57.3% [52.7, 61.9]	48.3%	77.5%	45.8%	57.5%
GPT-5-mini-Thinking	69.6% [65.4, 73.5]	85.8%	85.0%	60.8%	46.7%
Gemini 3.1 Pro-Thinking	74.8% [70.8, 78.8]	94.2%	93.3%	63.3%	48.3%
Claude Opus 4.5-Thinking	60.2% [55.8, 64.6]	74.2%	71.7%	45.0%	50.0%



Model	Studio ($n=158$)	Furnished ($n=152$)	Outdoor ($n=170$)
Qwen2.5-VL-7B-Instruct	63.9% (101)	63.8% (97)	64.1% (109)
Qwen3-VL-8B-Thinking	62.0% (98)	59.9% (91)	59.4% (101)
GLM-4.5V-Thinking	71.5% (113)	66.4% (101)	69.4% (118)
GPT-4o-mini	56.3% (89)	66.4% (101)	50.0% (85)
GPT-5-mini-Thinking	64.6% (102)	78.9% (120)	65.9% (112)
Gemini 3.1 Pro-Thinking	70.3% (111)	84.2% (128)	70.6% (120)
Claude Opus 4.5-Thinking	61.4% (97)	63.8% (97)	55.9% (95)

§2 Positional Biases

Model	Parse	Pred. L	Gold L
Qwen2.5-VL-7B-Instruct	100.0%	64.4%	53.8%
Qwen3-VL-8B-Thinking	100.0%	73.3%	53.8%
GLM-4.5V-Thinking	99.8%	49.0%	53.8%
GPT-4o-mini	100.0%	59.0%	53.8%
GPT-5-mini-Thinking	100.0%	40.0%	53.8%
Gemini 3.1 Pro-Thinking	100.0%	49.8%	53.8%
Claude Opus 4.5-Thinking	100.0%	44.4%	53.8%



“Left” inside or outside the scene?

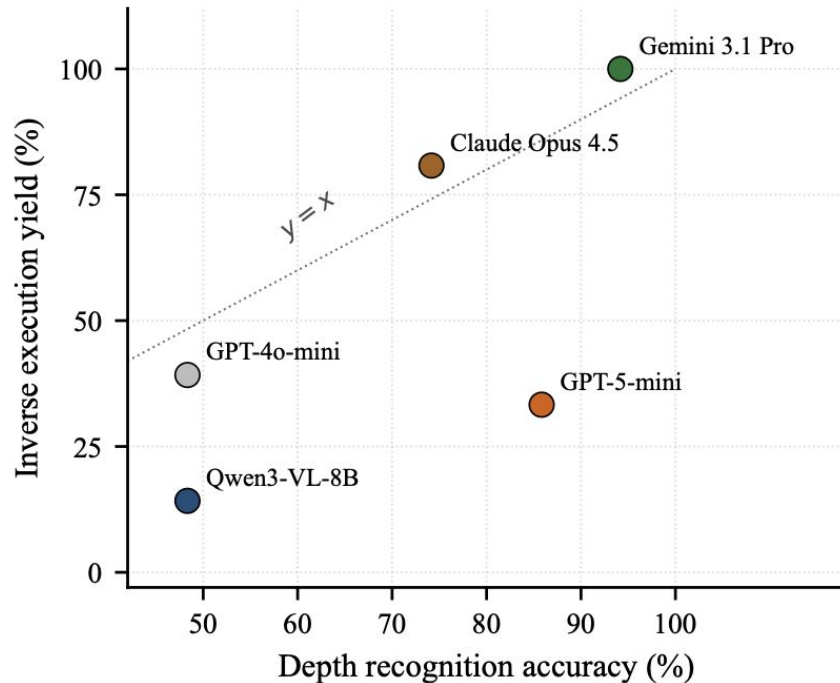
Same prompt, two valid frames of reference.



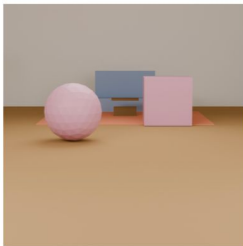
Results 2 — Inverse Generation

Oh no Qwen ;(Recognition does not transfer to Blender code generation.

Model	Executable	Rate
Gemini 3.1 Pro	120 / 120	100.0 %
Claude Opus 4.5	97 / 120	80.8 %
GPT-4o-mini	47 / 120	39.2 %
GPT-5-mini	40 / 120	33.3 %
Qwen3-VL-8B	17 / 120	14.2 %



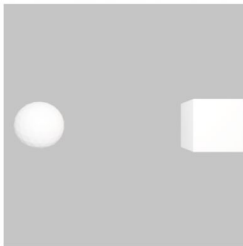
Reference



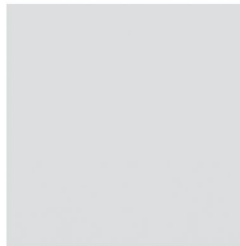
Gemini 3.1 Pro



Qwen3-VL-8B



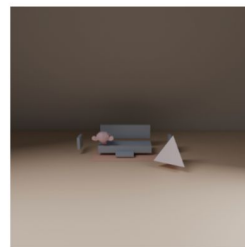
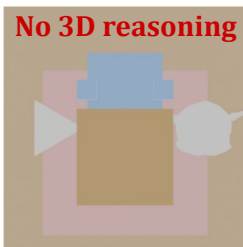
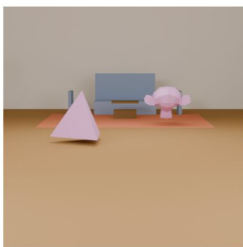
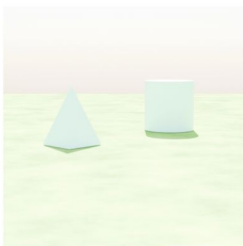
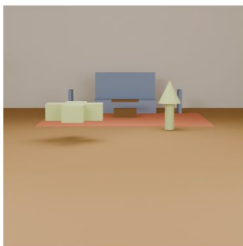
GPT-5-mini



Claude Opus 4.5



10



Conclusions

Graphics haven't been solved by VLMs — yet.

GRAFIQ measures the gap from object recognition to graphic understanding to inverse generation.

LESSONS LEARNED

Verification & prompts matter.

- Dropped 12% of ambiguous data.
- Iterated prompts to ensure clarity.

MAIN TAKEAWAYS

Two clear gaps remain.

- Gap 1 — object recognition → graphic understanding.
- Gap 2 — graphic reasoning → inverse generation.
- Frontier closed-source models lead, but not by enough.

FUTURE WORK

Next project ;)

- More diverse scenes, harder layouts.
- More tasks: shadows, occlusion, multi-view, physics.



GRAFIQ

Thanks for Listening!

Steve Chen · Patrick Tsung-Han Wu · Fei Gao · Alex Wang



SUPPLEMENTARY

Prompt Templates

depth_comparison **Depth Comparison**

1. *“From your viewpoint looking at this image, which object is closer to the camera — the one on your left or the one on your right?”*
2. *“In this image, which object is nearer to the viewer: the one on the left side of the picture or the one on the right side?”*
3. *“Looking at the image as you see it, which of the two objects appears at a shorter distance from the camera — the left one or the right one?”*

roughness_comparison **Roughness Comparison**

4. *“Looking at this image, which object has a shinier, more specular surface — the one on your left or the one on your right?”*
5. *“In the image as you see it, which object looks more glossy or mirror-like: the one on the left or the one on the right?”*
6. *“From your viewpoint, which of the two objects has a lower surface roughness (appears more reflective) — the left one or the right one?”*

normal_orientation **Normal Orientation**

7. *“From your viewpoint looking at this image, which object is rotated more toward the camera — the one on your left or the one on your right?”*
8. *“In the image as you see it, which object shows more of its front face to the viewer: the left one or the right one?”*
9. *“Looking at the image, one object faces the camera directly; the other is turned sideways. Which is facing you — the one on your left or the one on your right?”*

light_direction **Light Direction**

10. *“Looking at this image, which object is more brightly illuminated by the key light — the one on your left or the one on your right?”*
11. *“In the image as you see it, which of the two objects receives more direct light: the one on the left or the one on the right?”*
12. *“From your viewpoint, one object is strongly lit while the other is mostly in shadow. Which object is more brightly lit — the left one or the right one?”*



SUPPLEMENTARY

Inverse Generation Prompts

system_prompt Shared System Prompt

“You are a 3D-graphics expert specialising in Blender 4.4 Python scripting. Your task is to write a Python script that reconstructs the 3D scene shown in an image, using the bpy module.”

Guidelines for the model:

1. Load objects from the asset library.
2. Normalise each object to fit a unit cube.
3. Place objects at the correct left / right positions.
4. Set up the camera (location, rotation, lens).
5. Set up three-point lighting (key, fill, rim).
6. Set up the background.

Output requirement. The model must return only the Python code block — no conversational text.

SETTING 1 Zero-shot Generation

The model receives the system prompt, the scene image, and a single user instruction:

“Write a complete Blender Python script to reconstruct this scene exactly as shown.”

SETTING 2 Code Completion

Imports and initial object loading are pre-supplied per item; the model finishes placement, camera, and lighting.

“Complete the following Blender Python script to reconstruct this scene...”

